4.6 Similarity and Transformations

Name _____ Period ____ Date _____

Geometry 4.6 · Student sheet

I can…

- perform a similarity transformation.
- describe a similarity transformation that maps one figure onto another.
- decide whether two figures are similar.

Vocabulary · fill in the missing word

Similarity transformation — a dilation or a _____ of rigid motions and dilations

Similar figures — figures that can be mapped onto each other by a similarity _____

Part 1 · Discovery: same shape, different size

Small: J(−2, 1), K(−1, 2), L(−1, 1). Large: J′(2, −2), K′(4, 0), L′(4, −2).

	JK	KL	JL	m∠L
Small				
Large				
Large ÷ small				—

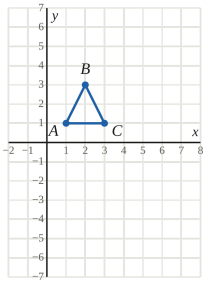
Moves that map the small triangle onto the large one: _____

Part 2 · Notes and examples (fill in each note, then work the example)

Similarity transformation: a dilation, or a composition of _____ and dilations.

Example 1

Dilate △ABC with A(1, 1), B(2, 3), C(3, 1) by a scale factor of 2 (center the origin), then reflect the image in the x-axis.



Example 2

Describe a similarity transformation that maps J(−2, 1), K(−1, 2), L(−1, 1) to J′(2, −2), K′(4, 0), L′(4, −2).

Similar figures: two figures are similar if and only if there is a _____ transformation that maps one onto the other.

Example 3

Are the rectangles similar?

a. 2 by 4 and 3 by 6

b. 2 by 4 and 4 by 6

What stays and what changes: angle measures stay the same; side lengths are multiplied by the _____.

Example 4

△ABC is dilated by a scale factor of 3. In △ABC, m∠A = 50° and AB = 7. Find m∠A′ and A′B′.

Example 5

Circle C has center (0, 0) and radius 2. Circle D has center (5, 1) and radius 6. Describe a similarity transformation that maps circle C onto circle D.

Part 3 · Partner practice

Partner 1

Dilate (2, −1) by a scale factor of 3 (center the origin), then translate by ⟨−4, 2⟩.

Partner 2

Is a square with side 3 similar to a square with side 5? If so, give the scale factor.